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Inventor(s)

Kavala; Asko et al.

Determining Reference Power Value for Estimating Saved Energy

Abstract

A method including collecting operational data associated with a motor controlled by a variable speed drive, wherein the operational data includes at least speed values and power values of the motor; determining, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimating, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.

Inventors: Kavala; Asko (Lappeenranta, FI), Pekkala; Asko (Helsinki, FI), Alkkiomäki; Olli (Helsinki, FI)

Applicant: ABB Schweiz AG (Baden, CH)

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Background/Summary

TECHNICAL FIELD

[0001] The following example embodiments relate to variable speed drives.

BACKGROUND

[0002] Currently, there is a heightened awareness regarding energy consumption and its impact on the environment, particularly concerning carbon dioxide emissions. Modern electronic devices are increasingly equipped with capabilities to monitor and calculate their own energy consumption as an embedded functionality. This monitored data can be utilized to enhance the efficiency of a system or production process.

[0003] The scope of protection sought for various example embodiments is set out by the claims. The example embodiments and features, if any, described in this specification that do not fall under the scope of the claims are to be interpreted as examples useful for understanding various embodiments.

SUMMARY

[0004] According to an aspect, there is provided an apparatus comprising at least one processor, and at least one memory storing instructions that, when executed by the at least one processor, cause the apparatus at least to: collect operational data associated with a motor controlled by a variable speed drive, wherein the operational data comprises at least speed values and power values of the motor; determine, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimate, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.

[0005] According to another aspect, there is provided a method comprising: collecting operational data associated with a motor controlled by a variable speed drive, wherein the operational data comprises at least speed values and power values of the motor; determining, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimating, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.

[0006] According to another aspect, there is provided a non-transitory computer readable medium comprising instructions which, when executed by an apparatus, cause the apparatus to perform at least the following: collecting operational data associated with a motor controlled by a variable speed drive, wherein the operational data comprises at least speed values and power values of the motor; determining, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimating, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In the following, various example embodiments will be described in greater detail with reference to the accompanying drawings, in which

[0008] FIG. 1 illustrates an example of a communication system;

[0009] FIG. 2 illustrates a signal flow diagram;

[0010] FIG. 3 illustrates a flow chart;

[0011] FIG. 4 illustrates a flow chart;

[0012] FIG. 5A illustrates an example of operational data;

[0013] FIG. 5B illustrates an example of generating a curve;

[0014] FIG. 5C illustrates an example of identifying a reference power value;

[0015] FIG. 6 illustrates an example of the error between a nominal power value and the reference power value; and

[0016] FIG. 7 illustrates an example of an apparatus.

DETAILED DESCRIPTION

[0017] The following embodiments are exemplifying. Although the specification may refer to “an”, “one”, or “some” embodiment(s) in several locations of the text, this does not necessarily mean that each reference is made to the same embodiment(s), or that a particular feature only applies to a single embodiment. Single features of different embodiments may also be combined to provide other embodiments. Furthermore, the words “comprising” and “including” should be understood as not limiting the described embodiments to consist of only those features that have been mentioned, and such embodiments may also contain features that have not been specifically mentioned.

Reference numbers, in the description and/or in the claims, serve to illustrate the embodiments with reference to the drawings, without limiting the embodiments to these examples only.

[0018] Currently, there is a heightened awareness regarding energy consumption and its impact on the environment, particularly concerning carbon dioxide emissions. Modern electronic devices are increasingly equipped with capabilities to monitor and calculate their own energy consumption as an embedded functionality. This monitored data can be utilized to enhance the efficiency of a system or production process, or to evaluate the merits of recently undertaken investments.

[0019] For example, in pump applications, the use of a variable speed drive (VSD) may be a significant energy saver when compared to conventional systems, which comprise throttling control paired with a direct on line (DOL) starter, especially in scenarios involving quadratic torque loads. In DOL motor control, the DOL motor starter connects the electric motor directly to the line voltage. DOL starters operate as a basic stop/start switch for an electric motor and may be fitted with overload protection devices to prevent short circuits and overheating.

[0020] The primary difference between DOL motor control and VSDs is that

[0021] DOL starters provide full voltage for direct motor start-up, resulting in high inrush current and fixed-speed operation, whereas VSDs gradually vary the motor's speed and torque by adjusting the power supply's frequency and voltage, offering greater control and efficiency.

[0022] For calculating the energy savings provided by variable speed drive applications, a reference motor power value under DOL control is needed. This reference motor power value represents the power that is needed to run the process or actuator in accordance with the torque requirement. In a fan or pump application with VSD control, the reference (nominal) speed of the motor typically results in a reduced power requirement to maintain the torque, as compared to DOL control. The difference in the required power between DOL and VSD control may be used as a basis for calculating the energy savings provided by VSD. Conventionally, the reference power value for DOL has been assumed to remain constant across the motor's speed range (from zero to its maximum speed).

[0023] The challenge herein lies in accurately defining the reference power value for the purpose of calculating the energy savings. While the nominal (nameplate) motor power may serve as a potential reference, variations in motor and actuator sizing may lead to significant deviations from this nominal (nameplate) value. For example, using the nominal power value of an oversized motor as a reference in the energy-saving calculations may result in overestimated energy savings. Typically, the range of motor operation points under VSD control, in terms of speed, is below the asynchronous motor's nominal speed. Therefore, specialized setups and measurements may be needed to determine the correct reference power value.

[0024] The definition of the reference power value can be set during the initial commissioning of the variable speed drive. However, usually there is very little operational data available at this stage. In practice, the motor's nominal (nameplate) power value is often employed as the reference for the energy savings calculation. Because of this uncertainty regarding the reference power value, the energy saving monitoring features of VSDs are seldom used.

[0025] The advent of cloud technology provides the capability to monitor variable speed drives and associated applications remotely. Given the increasing importance of energy consumption and savings, improved accuracy in calculating the energy savings is essential to sustainability. Some example embodiments provide a method for determining a reference power value in order to accurately estimate the amount of energy saved by a variable speed drive in comparison to DOL motor control. In some example embodiments, this reference power value may be determined based on remotely collected operational data of the variable speed drive. Thus, there is no need to perform any additional measurement actions on-site. Accurate estimation of the energy savings may be beneficial for optimizing energy efficiency and reducing environmental impact.

[0026] Different embodiments and examples are described below using single units, models, equipment and memory, without restricting the embodiments and examples to such a solution. Concepts called cloud computing and/or virtualization may be used. Virtualization may allow a single physical computing device to host one or more instances of virtual machines that appear and operate as independent computing devices, so that a single physical computing device can create, maintain, delete, or otherwise manage virtual machines in a dynamic manner. It is also possible that device operations will be distributed among a plurality of servers, nodes, devices or hosts. In cloud computing network devices, computing devices and/or storage devices provide shared resources. Some other technology advancements, such as Software-Defined Networking (SDN) may cause one or more of the functionalities described below to be migrated to any corresponding abstraction or apparatus or device. Therefore, all words and expressions should be interpreted broadly, and they are intended to illustrate, not to restrict, the example embodiments.

[0027] FIG. 1 illustrates an example of a communication system to which some example embodiments may be applied. Referring to FIG. 1, some example embodiments may be based on wireless or wired communications, such as 3G (third generation), 4G (fourth generation), LTE (long term evolution), LTE-A (long term evolution advanced), 5G (fifth generation), 5G NR (new radio), 6G (sixth generation), UMTS (universal mobile telecommunications system), EDGE (enhanced data rates for GSM evolution), WCDMA (wideband code division multiple access), Bluetooth, WLAN (wireless local area network), Wi-Fi, Li-Fi (light fidelity), Ethernet, or any other mobile or wireless or wired network. The communication may also occur between nodes belonging to different but compatible systems, such as LTE and 5G.

[0028] It should be noted that FIG. 1 illustrates a simplified system architecture only showing some elements and functional entities, all being logical units whose implementation may differ from what is shown. The connections shown in FIG. 1 are logical connections; the actual physical connections may be different. Data collection may use so-called master protocols, in which a master network node subscribes data from slaves (devices whose data it wants to have), and a slave device or network node sends its data to the receiver/master based on query or automatically based on subscription. It is apparent to a person skilled in the art that the system may also comprise other functions and structures. It should be appreciated that the functions, structures, elements, and protocols used in or for communication are irrelevant to the example embodiments. Therefore, they need not be discussed in more detail here.

[0029] Referring to FIG. 1, the system may comprise at least one variable speed drive **101**, at least one motor **102**, one or more sensor devices **103**, a gateway device **106**, a cloud platform **104**, and a user device **105**.

[0030] The variable speed drive **101** may also be referred to as a variable frequency drive. The variable speed drive **101** may be used to run or rotate an electric motor **102** at different speeds, wherein the electric motor **102** may run a pump or any other rotating equipment (e.g., fan, gearbox, belt drive, etc.). The variable speed drive **101** may be electrically connected to the electric motor **102**. The variable speed drive **101** may comprise or be connected to a controller, for example a proportional-integral-derivative (PID) controller. The controller may be configured to send control signals to the variable speed drive **101**. The variable speed drive **101** may control highly dynamic

industrial processes, in which for example the speed or torque applied to the motor **102** has to be varied according to the needs of the industrial process.

[0031] The variable speed drive **101** may be equipped with a short-range communication interface, such as Bluetooth, Ethernet, ZigBee, Li-Fi, Wi-Fi, wireless mesh network, near field communication (NFC), or any other wireless or wired connection. The short-range communication interface may be comprised, for example, in the variable speed drive **101** or in a control panel of the variable speed drive **101**. The variable speed drive **101** may be configured to communicate with the gateway device **106**, the motor **102** and/or the one or more sensor devices **103** via the short-range communication interface.

[0032] The one or more sensor devices **103** may be configured to measure operational data associated with the motor **102**. The operational data may comprise values of one or more variables associated with the motor **102**. For example, the one or more sensor devices **103** may measure values of at least one of: speed, torque, vibration, temperature, magnetic flux density, and/or electrical current of the motor **102** over time. There may be separate sensor devices for measuring each variable (e.g., separate sensors for vibration and temperature, respectively), or the variables may be measured by the same sensor device. The one or more sensor devices **103** may be comprised in the motor **102**, or attached to the motor **102**, or in close proximity to the motor **102**.

[0033] The variable speed drive **101** may be configured to estimate or calculate the speed and power output of the motor **102**. For example, the variable speed drive **101** may use one or more algorithms to estimate the motor's speed based on electrical parameters such as voltage, current, and the motor's electrical characteristics. In this way, the variable speed drive **101** may control the speed of the motor **102** without the need for a physical speed sensor. The power output of the motor **102** may be calculated by the variable speed drive **101** based on the voltage and current supplied to the motor **102**, and the motor's efficiency and power factor.

[0034] The variable speed drive **101** may be configured to communicate with the cloud platform **104** via the gateway device **106**. For example, the variable speed drive **101** may transmit the operational data to the cloud platform **104** via the gateway device **106**. The gateway device **106** may comprise, for example, an edge gateway or a user device (which may be the same as the user device **105**, or different from the user device **105**). For example, a mobile phone (or any other user device) may be used as the gateway device **106** to push the operational data to the cloud platform **104** via an internet connection. The user device used as the gateway device **106** may also be used to view historical measurement information from the one or more sensor devices **103** (stored at the cloud platform **104**) via the internet connection.

[0035] For enabling the connection between the variable speed drive **101** and the gateway device **106**, the gateway device **106** may also be equipped with a short-range communication interface, such as Bluetooth, Ethernet, ZigBee, Li-Fi, Wi-Fi, wireless mesh network, NFC, or any other wireless or wired connection.

[0036] The gateway device **106** may be connected to the internet via a network interface, such as 3G, 4G, LTE, LTE-A, 5G, 5G NR, 6G, UMTS, EDGE, WCDMA, WLAN, Wi-Fi, Li-Fi, Ethernet, or any other mobile, wireless or wired network. The gateway device **106** may be connected to the cloud platform **104** via the network interface.

[0037] The cloud platform **104**, also known as a cloud computing platform, refers to a comprehensive and integrated set of cloud-based services and resources that enable users to build, deploy, manage, and scale various applications and services over the internet. It provides a flexible and scalable infrastructure for hosting applications, storing data, and executing computing tasks without the need for on-premises hardware and infrastructure.

[0038] The cloud platform **104** may be configured to collect or receive the operational data from the variable speed drive **101** via the gateway device **106**. The cloud platform **104** may further be configured to store the received operational data in at least one memory or database.

[0039] The cloud platform **104** or the variable speed drive **101** may be configured to determine,

based on the operational data, a reference power value corresponding to a nominal speed value of the motor **102**; and estimate, based on the reference power value, an amount of energy saved by the variable speed drive **101** compared to direct on line motor control.

[0040] The cloud platform **104** may further be connected to a user device **105** to enable a user of the user device **105** to monitor the condition of the motor **102** via a user interface based on the operational data. The cloud platform **104** may be configured to indicate, to the user device **105**, the amount of energy saved.

[0041] The user device **105** may comprise user equipment such as a smartphone, mobile phone, tablet computer, laptop computer, desktop computer, or any other computing device. The user device **105** may be a remote device located at a different site than the motor **102**. Alternatively, the user device **105** may be a local device located on-site at a close proximity to the motor **102**.

[0042] The user device **105** may be connected to the internet via a network interface, such as 3G, 4G, LTE, LTE-A, 5G, 5G NR, 6G, UMTS, EDGE, WCDMA, WLAN, Wi-Fi, Li-Fi, Ethernet, or any other mobile, wireless or wired network. The user device **105** may be configured to exchange information, i.e., to transmit and/or receive data, with the cloud platform **104** via the network interface.

[0043] FIG. 2 illustrates a signal flow diagram according to an example embodiment.

[0044] Referring to FIG. 2, at **201**, the cloud platform **104** collects or receives, from the variable speed drive **101**, operational data associated with the motor **102** controlled by the variable speed drive **101**, wherein the operational data comprises at least speed values and power values of the motor **102**. The speed values (speed output) and power values (power output) of the motor **102** may be measured during the operation of the motor **102** and transmitted to the cloud platform **104** in real time or at certain time intervals.

[0045] The operational data may be collected over a pre-determined time period, wherein the length of this pre-determined time period depends on the process variations of the target application. For example, the motor **102** may be part of a pump application or a fan application. The pre-determined time period may be, for example, at least a day, a week, or another suitable duration.

[0046] In pump or fan applications, the operational speed of the motor **102** typically remains below the field weakening point of the motor **102** (i.e., a point where the motor's magnetic field becomes saturated, leading to a decrease in torque). The controlled speed values may be within a relatively narrow range due to the efficiency characteristics of the actuator (i.e., the device that converts the motor's rotational motion into useful mechanical work).

[0047] At **202**, the cloud platform **104** generates, based on the operational data, a curve representing a relationship between the speed values and the power values of the motor **102**, starting from zero speed. In other words, the curve indicates how the motor **102** performs at different speeds. An example of the curve **510** is shown in FIG. 5B and FIG. 5C.

[0048] The curve may be represented by a third-degree polynomial function $P_{sub.VSD}(n) \approx a n_{sup.3} + b n_{sup.2} + c n + d$, where a, b, c, and d are adjusting constants. The curve may be fitted to the collected operational data by using any suitable computational software.

[0049] In this way, a model may be created for defining a reference power value for estimating the amount of energy saved by the variable speed drive **101**.

[0050] At **203**, the cloud platform **104** determines or identifies the reference power value according to a point on the curve corresponding to a nominal speed value of the motor **102**. An example of identifying the reference power value **523** on the curve **510** based on the nominal speed value **521** is shown in FIG. 5C. The nominal speed value of the motor **102** is information that is available at the variable speed drive **101**, and the cloud platform **104** may obtain this information from the variable speed drive **101** (e.g., by reading the nominal speed value from the parameter setup of the variable speed drive **101**).

[0051] In other words, at the motor's nominal speed (i.e., the speed at which it is designed to

operate most efficiently, often indicated on its nameplate), a specific operation point is defined. This operation point represents the motor's power output under direct on line motor control. This operation point serves as the reference power value for calculating the energy savings of the variable speed drive **101** compared to direct on line motor control.

[0052] At **204**, the cloud platform **104** estimates, based on the reference power value, an amount of energy saved by the variable speed drive **101** compared to direct on line motor control. In other words, the reference power value may be used as a reference for estimating the amount of energy saved.

[0053] The energy consumption of the motor **102** under DOL motor control can be estimated, for example, using the formula: $E_{\text{sub.DOL}} = P_{\text{sub.ref}} * T$, where $E_{\text{sub.DOL}}$ is the amount of energy consumed by the motor **102** under DOL control (e.g., in kilowatt-hours), $P_{\text{sub.ref}}$ is the reference power value (e.g., in kilowatts), and T is the operating time of the motor **102** (e.g., in hours or days).

[0054] The variable speed drive **101** adjusts the speed and load of the motor **102**, which results in different energy consumption patterns. The energy consumed by the motor **102** under VSD control can be approximated using the Affinity Laws, which suggest that power varies with the cube of the speed for centrifugal loads. For example, the energy consumed by the motor **102** under

$$[00001] E_{\text{VSD}} = P_{\text{ref}} * T * LF * \left(\frac{N_{\text{VSD}}}{N_{\text{rated}}} \right)^3,$$

where $E_{\text{sub.VSD}}$ is the amount of energy consumed by the motor **102** under VSD control (e.g., in kilowatt-hours), $P_{\text{sub.ref}}$ is the reference power value (e.g., in kilowatts), T is the operating time of the motor **102** (e.g., in hours or days), LF is the load factor (i.e., the ratio of actual load to rated load), $N_{\text{sub.VSD}}$ is the average speed of the motor **102** with VSD, and $N_{\text{sub.rated}}$ is the rated speed of the motor **102**. The $P_{\text{sub.ref}}$ value refers to the motor power when connected directly to the supply. The $P_{\text{sub.ref}}$ value may be used for reference when energy savings are calculated. The accuracy of the energy savings calculation is directly dependent on the accuracy of the $P_{\text{sub.ref}}$ value.

[0055] The amount of energy saved may then be calculated as the difference in the energy consumption of the motor **102** in DOL and VSD operation: $\text{Energy Savings} = E_{\text{sub.DOL}} - E_{\text{sub.VSD}}$

[0056] The calculation of the amount of energy saved may have various purposes. For example, the calculation of the amount of energy saved may be used to prove the energy saving effect of the selected target application (e.g., compared to similar assets and/or fleets), or to enable an energy consumption review of the selected target application (e.g., in terms of energy usage, actual load factor, or oversizing or undersizing of the motor **102** or the actuator). As another example, the calculation of the amount of energy saved may be used to detect from the application that an advanced energy optimization function may increase the efficiency of the powertrain. Moreover, trusting in the result of energy saving calculation may enable more energy-improving concepts.

[0057] At **205**, the cloud platform **104** may indicate, to the user device **105**, the amount of energy saved.

[0058] At **206**, the user device **105** may display the amount of energy saved to a user, for example via a graphical user interface.

[0059] In an alternative embodiment, the amount of energy saved may be estimated as an embedded functionality of the variable speed drive **101** (instead of the cloud **104**). This is described in the following with reference to FIG. 3.

[0060] FIG. 3 illustrates a flow chart according to an example embodiment of a method performed by a variable speed drive **101**.

[0061] Referring to FIG. 3, in block **301**, the variable speed drive **101** collects operational data associated with a motor **102** controlled by the variable speed drive **101**, wherein the operational data comprises at least speed values and power values of the motor **102**.

[0062] The operational data may be collected over a pre-determined time period sufficient to represent process variations in a target application. For example, the motor **102** may be part of a pump application or a fan application.

[0063] In block **302**, the variable speed drive **101** generates, based on the operational data, a curve representing a relationship between the speed values and the power values of the motor **102**.

[0064] The curve may be represented by a third-degree polynomial function

$P_{sub.VSD}(n) \approx a n_{sup.3} + b n_{sup.2} + c n + d$, where a, b, c, and d are adjusting constants. The curve may be fitted to the collected operational data by using any suitable computational software.

[0065] In block **303**, the variable speed drive **101** determines or identifies a reference power value according to a point on the curve corresponding to a nominal speed value of the motor **102**.

[0066] In block **304**, the variable speed drive **101** estimates, based on the reference power value, an amount of energy saved by the variable speed drive **101** compared to direct on line motor control. In other words, the reference power value may be used as a reference for estimating the amount of energy saved.

[0067] In block **305**, the variable speed drive **101** indicates, to a user, the amount of energy saved (e.g., via a user interface).

[0068] The variable speed drive **101** may determine one or more control parameter changes (e.g., optimizing flux control) for the variable speed drive **101** based on the amount of energy saved, and propose the one or more control parameter changes to the user (e.g., via the user interface).

[0069] FIG. **4** illustrates a flow chart according to an example embodiment of a method (e.g., a computer-implemented method) for estimating energy savings. The method of FIG. **4** may be performed by one or more cloud services of the cloud platform **104**, or an apparatus **700** comprising, or comprised in, the cloud platform **104**, or the variable speed drive **101** or any other computing device.

[0070] Referring to FIG. **4**, in block **401**, the apparatus **700** collects operational data associated with a motor **102** controlled by the variable speed drive **101**, wherein the operational data comprises at least speed values and power values of the motor **102**. The motor **102** may be an electric motor.

[0071] The operational data may be collected over a pre-determined time period sufficient to represent process variations in a target application. For example, the target application may comprise a pump application or a fan application that the motor **102** may be part of.

[0072] In block **402**, the apparatus **700** determines, based on the operational data, a reference power value corresponding to a nominal speed value of the motor **102**.

[0073] In block **403**, the apparatus **700** estimates, based on the reference power value, an amount of energy saved by the variable speed drive **101** compared to direct on line motor control. In other words, the reference power value may be used as a reference for estimating the amount of energy saved.

[0074] The apparatus **700** may generate, based on the operational data, a curve representing a relationship between the speed values and the power values of the motor **102**. The curve may be represented by a third-degree polynomial function.

[0075] The reference power value may be identified according to a point on the curve corresponding to the nominal speed value of the motor **102**.

[0076] The apparatus **700** may indicate, to a user, the amount of energy saved.

[0077] The functions and/or blocks described above by means of FIG. **2**, FIG. **3** and FIG. **4** are in no absolute chronological order, and some of them may be performed simultaneously or in an order differing from the described one. Other functions and/or blocks may also be executed between them or within them. Some of the functions and/or blocks can also be left out or replaced by a corresponding function or block.

[0078] As used herein, “at least one of the following: <a list of two or more elements>” and “at least one of <a list of two or more elements>” and similar wording, where the list of two or more

elements are joined by “and” or “or”, mean at least any one of the elements, or at least any two or more of the elements, or at least all the elements.

[0079] FIG. 5A illustrates an example of the operational data. In FIG. 5A, three motor speed and power points **501**, **502**, **503** of the selected application are illustrated in a scatter plot, where the vertical axis represents the motor power (in kilowatts), and the horizontal axis represents the motor speed (in rotations per minute). In other words, each point **501**, **502**, **503** represents a pair of power and speed values. In practice, this is a set of measurement points from a monitored time period.

[0080] FIG. 5B illustrates an example of generating the curve **510**. In FIG. 5B, the curve **510** is fitted to follow the operation points **501**, **502**, **503**. It is not necessary to have operation points at the maximum power of the motor **102**. The loading function representing the curve can be formed automatically using suitable computational software.

[0081] FIG. 5C illustrates an example of identifying the reference power value **523** according to a point **522** on the curve **510** corresponding to the nominal speed value **521** of the motor **102**. In FIG. 5C, a vertical line at the nominal speed **521** of the motor **102** has been drawn. This line crosses the curve **510** at the point **522**, which represents the needed power for the DOL motor control.

Respectively, this reference power value **523** can be calculated automatically.

[0082] FIG. 6 illustrates an example of the error **601** between the nominal (nameplate) power value **602** of the motor **102** and the reference power value **523** defined according to the actual application. In FIG. 6, the area **603** visualizes the amount of energy saved, as estimated based on the reference power value **523**. Using the nominal power value **602** would result in overestimated energy savings. Thus, by using the reference power value **523** for estimating the energy savings, the error in the energy savings estimation may be reduced in comparison to using the nominal power value **602** for estimating the energy savings.

[0083] FIG. 7 illustrates an example of an apparatus **700** comprising means for performing one or more of the example embodiments described above. The apparatus **700** may comprise, or be comprised in, the cloud platform **104** or a server of the cloud platform **104**. For example, the means may comprise one or more cloud services of the cloud platform **104**, or the means may comprise virtualization infrastructure. Cloud services refer to a wide range of on-demand computing resources and applications that may be provided to users over the internet by the cloud platform **104**. Alternatively, the apparatus **700** may comprise, or be comprised in, the variable speed drive **101** or any other computing device.

[0084] The apparatus **700** may comprise, for example, a circuitry or a chipset applicable for realizing one or more of the example embodiments described above. The apparatus **700** may be an electronic device or computing system comprising one or more electronic circuitries. The apparatus **700** may comprise a control circuitry **710** such as at least one processor, and at least one memory **720** storing instructions **722** which, when executed by the at least one processor, cause the apparatus **700** to carry out one or more of the example embodiments described above. Such instructions **722** may, for example, include computer program code (software). The at least one processor and the at least one memory storing the instructions may provide the means for providing or causing the performance of any of the methods and/or blocks described above.

[0085] The processor is coupled to the memory **720**. The processor is configured to read and write data to and from the memory **720**. The memory **720** may comprise one or more memory units. The memory units may be volatile or non-volatile. It is to be noted that there may be one or more units of non-volatile memory and one or more units of volatile memory or, alternatively, one or more units of non-volatile memory, or, alternatively, one or more units of volatile memory. Volatile memory may be for example random-access memory (RAM), dynamic random-access memory (DRAM) or synchronous dynamic random-access memory (SDRAM). Non-volatile memory may be for example read-only memory (ROM), programmable read-only memory (PROM), electronically erasable programmable read-only memory (EEPROM), flash memory, optical storage or magnetic storage. In general, memories may be referred to as non-transitory computer

readable media. The term “non-transitory,” as used herein, is a limitation of the medium itself (i.e., tangible, not a signal) as opposed to a limitation on data storage persistency (e.g., RAM vs. ROM). The memory **720** stores computer readable instructions that are executed by the processor. For example, non-volatile memory stores the computer readable instructions, and the processor executes the instructions using volatile memory for temporary storage of data and/or instructions. [0086] The computer readable instructions may have been pre-stored to the memory **720** or, alternatively or additionally, they may be received, by the apparatus, via an electromagnetic carrier signal and/or may be copied from a physical entity such as a computer program product. Execution of the computer readable instructions causes the apparatus **700** to perform one or more of the functionalities described above.

[0087] The memory **720** may be implemented using any suitable data storage technology, such as semiconductor-based memory devices, flash memory, magnetic memory devices and systems, optical memory devices and systems, fixed memory and/or removable memory.

[0088] The apparatus **700** may further comprise or be connected to a communication interface **730** comprising hardware and/or software for realizing communication connectivity according to one or more communication protocols. The communication interface **730** may comprise at least one transmitter and at least one receiver that may be integrated to the apparatus **700** or that the apparatus **700** may be connected to. The communication interface **730** may provide means for performing some of the blocks for one or more example embodiments described above. The communication interface **730** may comprise one or more components, such as: power amplifier, digital front end (DFE), analog-to-digital converter (ADC), digital-to-analog converter (DAC), frequency converter, (de)modulator, and/or encoder/decoder circuitries, controlled by the corresponding controlling units.

[0089] The communication interface **730** provides the apparatus with communication capabilities to communicate in the wireless communication network. The communication interface **730** may, for example, provide a network interface for communicating with the one or more sensor devices **103** and/or the gateway device **106**.

[0090] It is to be noted that the apparatus **700** may further comprise various components not illustrated in FIG. 7. The various components may be hardware components and/or software components.

[0091] As used in this application, the term “circuitry” may refer to one or more or all of the following: a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); and b) combinations of hardware circuits and software, such as (as applicable): i) a combination of analog and/or digital hardware circuit(s) with software/firmware and ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone, to perform various functions); and c) hardware circuit(s) and/or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (for example firmware) for operation, but the software may not be present when it is not needed for operation.

[0092] This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term circuitry also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.

[0093] The techniques and methods described herein may be implemented by various means. For example, these techniques may be implemented in hardware (one or more devices), firmware (one or more devices), software (one or more modules), or combinations thereof. For a hardware implementation, the apparatus(es) of example embodiments may be implemented within one or

more application-specific integrated circuits (ASICs), digital signal processors (DSPs), digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), graphics processing units (GPUs), processors, controllers, micro- controllers, microprocessors, other electronic units designed to perform the functions described herein, or a combination thereof. For firmware or software, the implementation can be carried out through modules of at least one chipset (for example procedures, functions, and so on) that perform the functions described herein. The software codes may be stored in a memory unit and executed by processors. The memory unit may be implemented within the processor or externally to the processor. In the latter case, it can be communicatively coupled to the processor via various means, as is known in the art. Additionally, the components of the systems described herein may be rearranged and/or complemented by additional components in order to facilitate the achievements of the various aspects, etc., described with regard thereto, and they are not limited to the precise configurations set forth in the given figures, as will be appreciated by one skilled in the art. [0094] It will be obvious to a person skilled in the art that, as technology advances, the inventive concept may be implemented in various ways within the scope of the claims. The embodiments are not limited to the example embodiments described above, but may vary within the scope of the claims. Therefore, all words and expressions should be interpreted broadly, and they are intended to illustrate, not to restrict, the embodiments.

Claims

1. An apparatus comprising at least one processor, and at least one memory storing instructions that, when executed by the at least one processor, cause the apparatus at least to: collect operational data associated with a motor controlled by a variable speed drive, wherein the operational data includes at least speed values and power values of the motor; determine, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimate, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.
2. The apparatus of claim 1, further being caused to: generate, based on the operational data, a curve representing a relationship between the speed values and the power values of the motor, wherein the reference power value is determined by identifying the reference power value according to a point on the curve corresponding to the nominal speed value of the motor.
3. The apparatus of claim 2, wherein the curve is represented by a third-degree polynomial function.
4. The apparatus of claim 1, wherein the operational data is collected over a pre-determined time period sufficient to represent process variations in a target application that the motor is part of.
5. The apparatus of claim 1, wherein the motor is part of a pump application or a fan application.
6. The apparatus of claim 1, wherein the amount of energy saved is estimated by using the reference power value as a reference for estimating the amount of energy saved.
7. The apparatus of claim 1, further being caused to: indicate, to a user, the amount of energy saved.
8. The apparatus of claim 1, wherein the apparatus includes, or is comprised in, the variable speed drive.
9. The apparatus of claim 1, wherein the apparatus includes, or is comprised in, a cloud platform.
10. A method comprising: collecting operational data associated with a motor controlled by a variable speed drive, wherein the operational data includes at least speed values and power values of the motor; determining, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimating, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.
11. The method of claim 10, further comprising: generating, based on the operational data, a curve

representing a relationship between the speed values and the power values of the motor, wherein the reference power value is determined by identifying the reference power value according to a point on the curve corresponding to the nominal speed value of the motor.

12. The method of claim 11, wherein the curve is represented by a third-degree polynomial function.

13. The method of claim 10, wherein the operational data is collected over a pre-determined time period sufficient to represent process variations in a target application that the motor is part of.

14. The method of claim 10, wherein the motor is part of a pump application or a fan application.

15. The method of claim 10, wherein the amount of energy saved is estimated by using the reference power value as a reference for estimating the amount of energy saved.

16. The method of claim 10, further comprising: indicating, to a user, the amount of energy saved.

17. The method of claim 10, wherein the method is performed by the variable speed drive.

18. The method of claim 10, wherein the method is performed by a cloud platform.

19. A non-transitory computer readable medium comprising instructions which, when executed by an apparatus, cause the apparatus to perform at least the following: collecting operational data associated with a motor controlled by a variable speed drive, wherein the operational data comprises at least speed values and power values of the motor; determining, based on the operational data, a reference power value corresponding to a nominal speed value of the motor; and estimating, based on the reference power value, an amount of energy saved by the variable speed drive compared to direct on line motor control.
